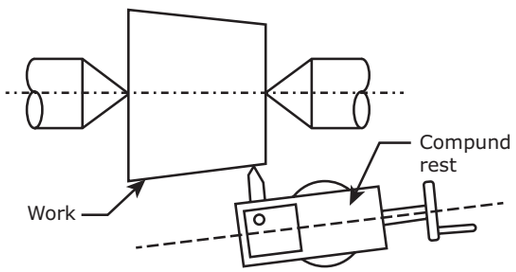
The compound is swiveled to the angle as above and clamped. Feed is given to the compound slide to generate the required taper.





Compound Rest Method

Note On doing taper turning in compound rest method, feed will be given to the tool from small diameter to large diameter.

Advantage

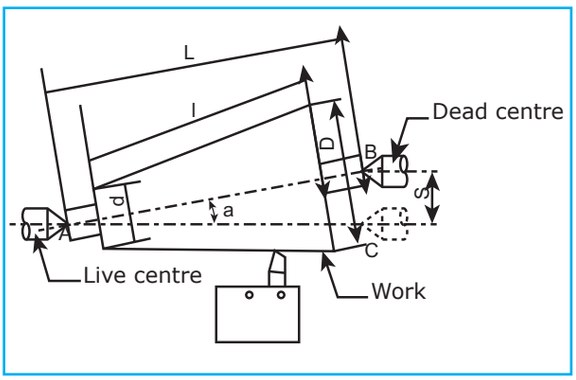
Taper turning can be done on outer surface of the workpiece and inner surface of the hole.

Disadvantage

1. This method is not suitable for long workpiece, because compound slide moves only small distance.
2. This method is done manually so, it takes more time and the productivity is less.

Tailstock setover method

Turning taper by the setover method is done by shifting the axis of rotation of the workpiece at an angle to the lathe axis and feeding the tool parallel to the lathe axis. The construction of tailstock is designed to have two parts namely the base and the body. The base is fitted on the bed guideways and the body having the dead centre can be moved at cross to shift the lathe axis.



The amount of setover-S, can be calculated as follows

$$S = \frac{D - d}{2} \times \frac{L}{l}$$

Where,

S – Amount of setover

D – Larger diameter

d – Smaller diameter

L – Length of the work

l – Length of the taper

The dead centre is suitably shifted from its original position to the calculated distance. The work is held between centres and longitudinal feed is given by the carriage to generate the taper.

Note While doing taper turning in a stepped turned work piece, each of the stepped taper turning must be calculated individually.

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